

10 Scalable Challenges in Combinatorics: A Public Benchmark Formalised in LEAN 4

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AI-Generated Math-Output

Artificial Intelligence (AI) can now solve or help solve difficult math problems. Recent progress by AI:

- Cycle Double Cover Conjecture (CDC), **proved** in July 2026
- Erdős' unit distance conjecture, **disproved** in May 2026

New claims require expert review for

1. Correctness
2. Significance

Keeping track of and verifying proposed solutions takes up time.

Idea

Experts of a field select a set of problems they consider significant and use them as a benchmark for AI progress in that field.

We designed TEN CHALLENGES:

- 1 10 significant and open combinatorial problems were selected.
- 2 Each challenge is formalised in LEAN 4 using Mathlib.
- 3 Our website checks submissions automatically.
- 4 Every formalisation is thoroughly checked to reduce errors.
- 5 Each challenge is parametrised by a number, building towards the full solution as the number grows.

The language: LEAN 4 and Mathlib

LEAN 4 is a computer language which checks mathematical problem statements and solutions automatically. A proposed proof either compiles or produces an error, so no referee is needed.

Mathlib is the community library which contains thousands of definitions and theorems, maintained by many contributors.

Hence we pose problems in LEAN 4:

```
def statement_N (r : ℕ) : Prop := ...      -- the problem, fixed by us
theorem challenge_N : statement_N r := ... -- the proof, supplied by the contestant
```

A submission is accepted exactly when LEAN 4 accepts the proof of the unchanged statement.

The ten challenges

Ten well-known open problems of combinatorics:

- ① Hadwiger conjecture
- ② Excluded minors of $\text{GF}(p^m)$ -representable matroids
- ③ Ramsey numbers
- ④ Sidorenko conjecture for bipartite graphs with bounded sides
- ⑤ Erdős–Hajnal conjecture for paths
- ⑥ Better-quasi-ordering of finite graphs under the minor relation
- ⑦ Sphere packing problem
- ⑧ Ryser's hypergraph conjecture
- ⑨ Union-closed sets conjecture
- ⑩ Unfriendly partition conjecture

They were selected for their significance and for admitting a scaling by a parameter r .

Example Statement: The Hadwiger Conjecture

Hadwiger Conjecture for r : *every graph with no K_{r+1} -minor is r -colourable.*

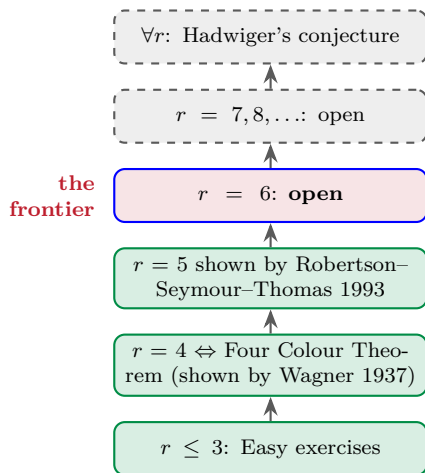
Challenge 1, Lean 4 formalization

```
/- The Hadwiger number: the largest r such that K_r is a minor of G. -/
noncomputable def hadwigerNumber {V : Type*} [Fintype V] (G : SimpleGraph V) : ℕ :=
  ...

def statement_01 (r : ℕ) : Prop :=
  ∀ {V : Type*} [Fintype V] (G : SimpleGraph V),
  hadwigerNumber G ≤ r → G.Colorable r

def r : ℕ := sorry -- the challenge parameter
theorem challenge_1 : statement_01 r := sorry
```

Growing difficulty with r



- Small r : an exercise or a hard theorem.
- The first open r : a genuine research problem, a solution is publishable.
- All r : the full conjecture.

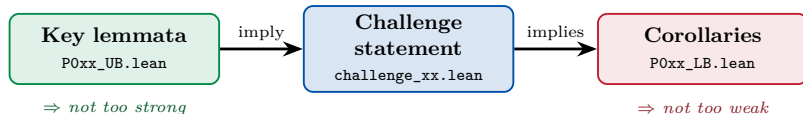
The risk

The LEAN 4 statement might silently differ from the intended mathematical meaning.

We apply the method of “Theorem Sandwich”, where we prove in LEAN 4:

- that the statement in question **follows from** known lemmata
- and that the statement **implies** interesting corollaries

This anchors the statement in established mathematics on both sides.



Example of an upper bound for challenge 1: The Hajos' Conjecture implies the Hadwiger Conjecture

The submission system

A contestant submits a LEAN 4 proof for a chosen parameter r , or a proof or disproof of the full statement.

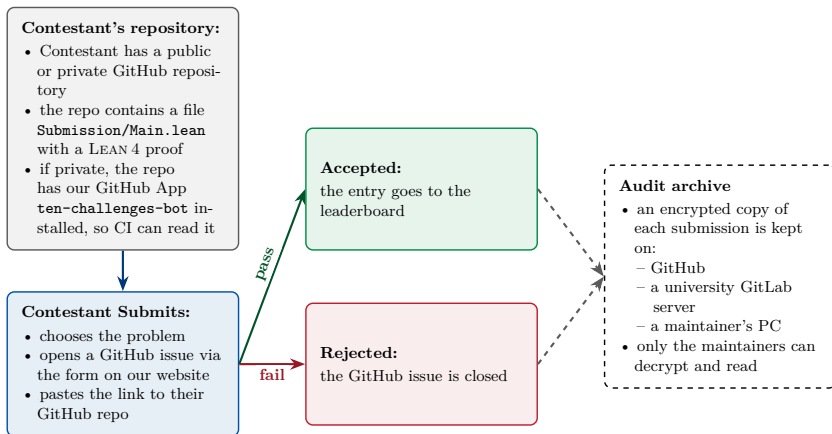
A submission

```
import Challenges.challenge_01
namespace Submission
def r : ℕ := 4
theorem challenge_1 : statement_01 r := by
...           -- the contestant's proof
end Submission
```

Checked automatically on our website:

- it must compile against the **unmodified** statement;
- no **sorry**; only Lean's standard axioms;
- accepted submissions go to a leaderboard with public and private tracks.

The submission pipeline



Conclusion

- 10 central open problems of combinatorics, formalised in LEAN 4, with automatic checking and a leaderboard.
- Parametrised: exercises at small r , known results near the frontier, then open problems — progress is visible before a conjecture is settled.
- A full solution or a solution for an open value r is a significant result.



[tenchallenges.github.io/
ten-challenges](https://tenchallenges.github.io/ten-challenges)

Thank you.

THANK YOU!



<https://tenchallenges.github.io/ten-challenges/>